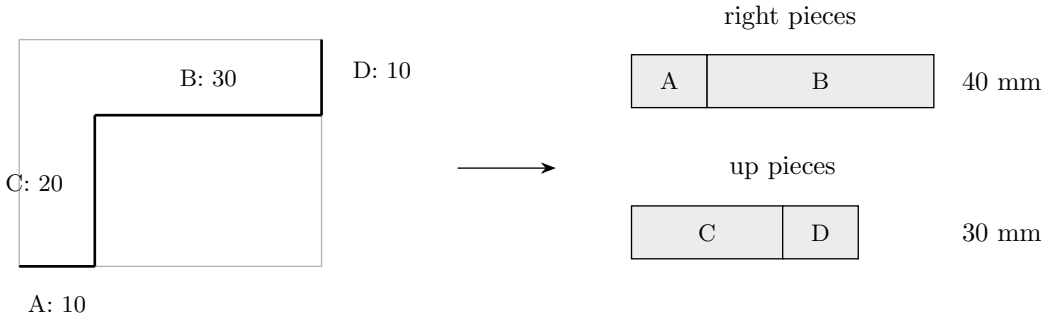
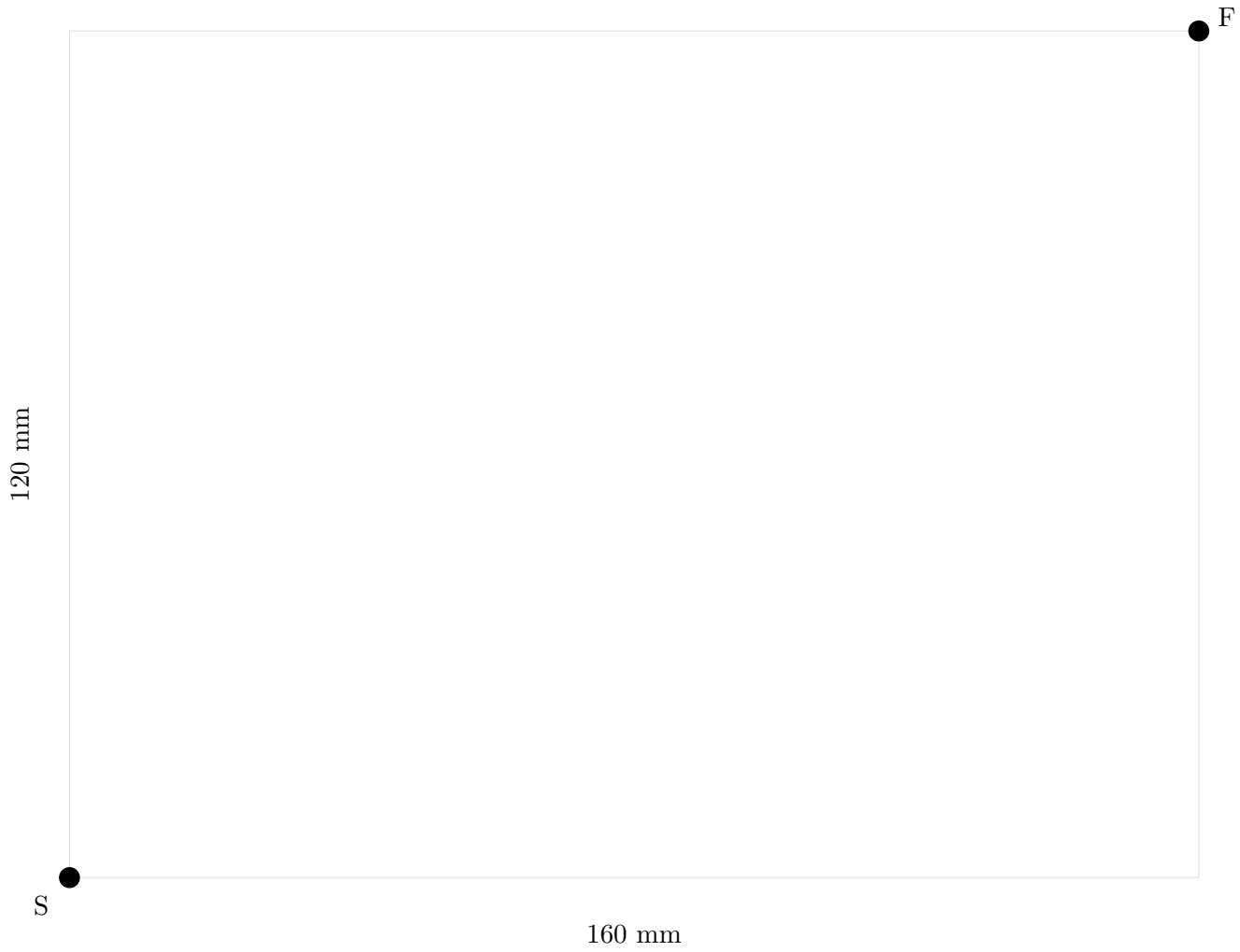
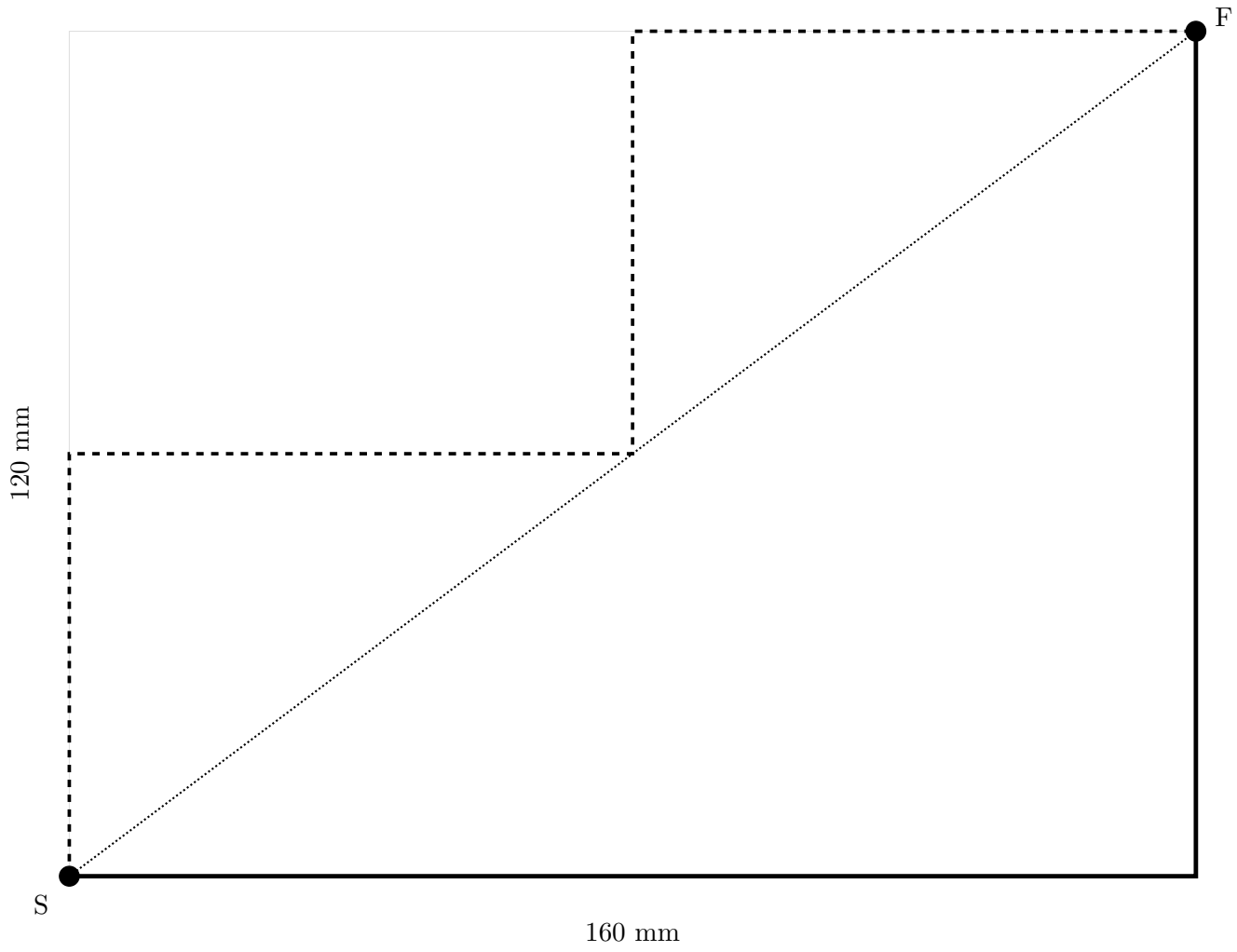
A staircase goes only right and up. Measure along all its pieces.



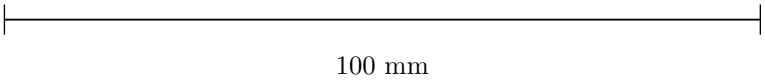
Problem 1: Make two different staircases from S to F. Use string to compare their lengths.



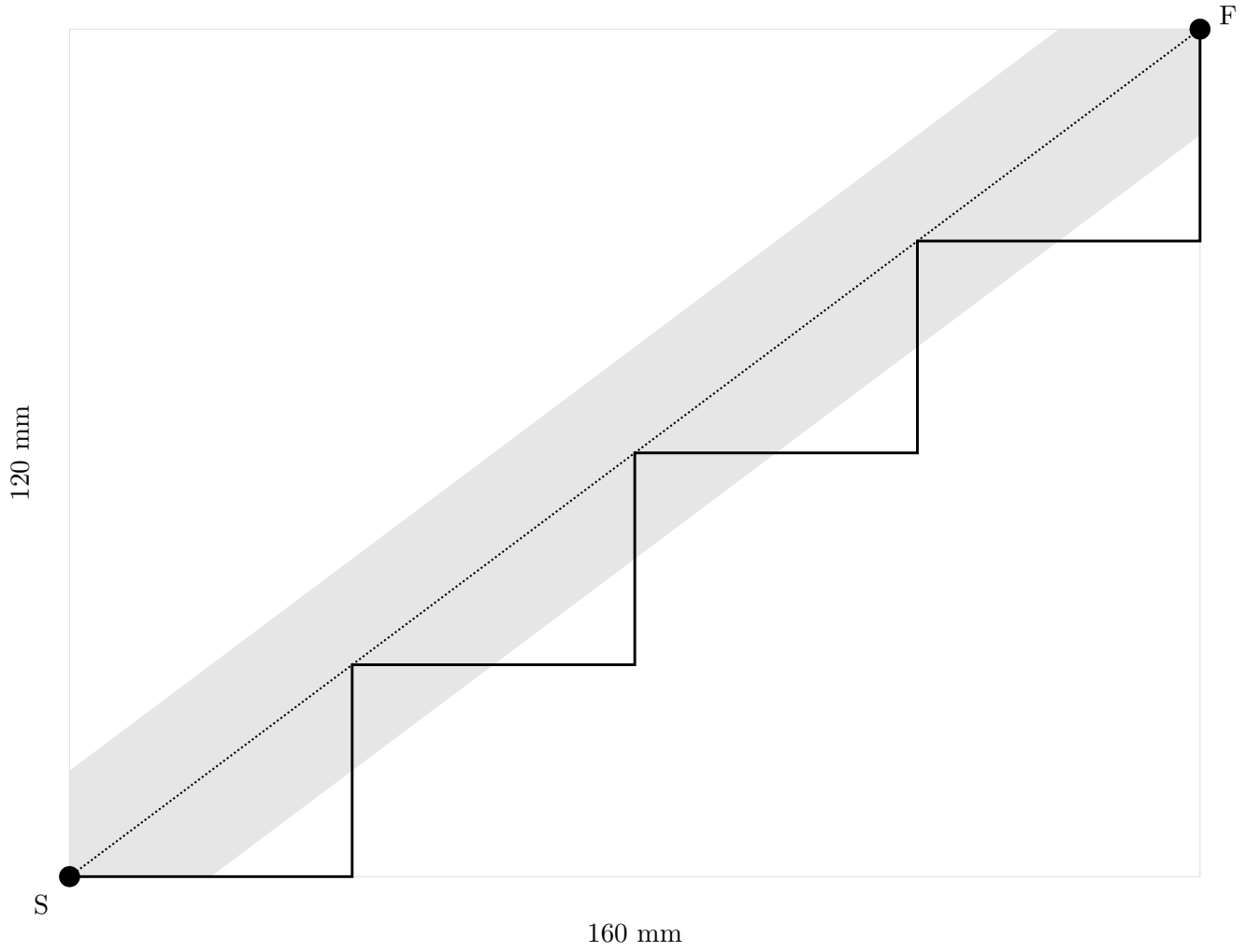
Problem 2: Compare the lengths of these two staircases. Find their exact lengths using the width and height.



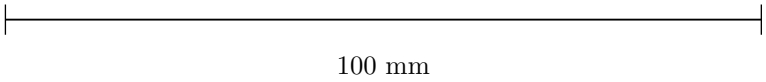
Solid: _____ mm Dashed: _____ mm



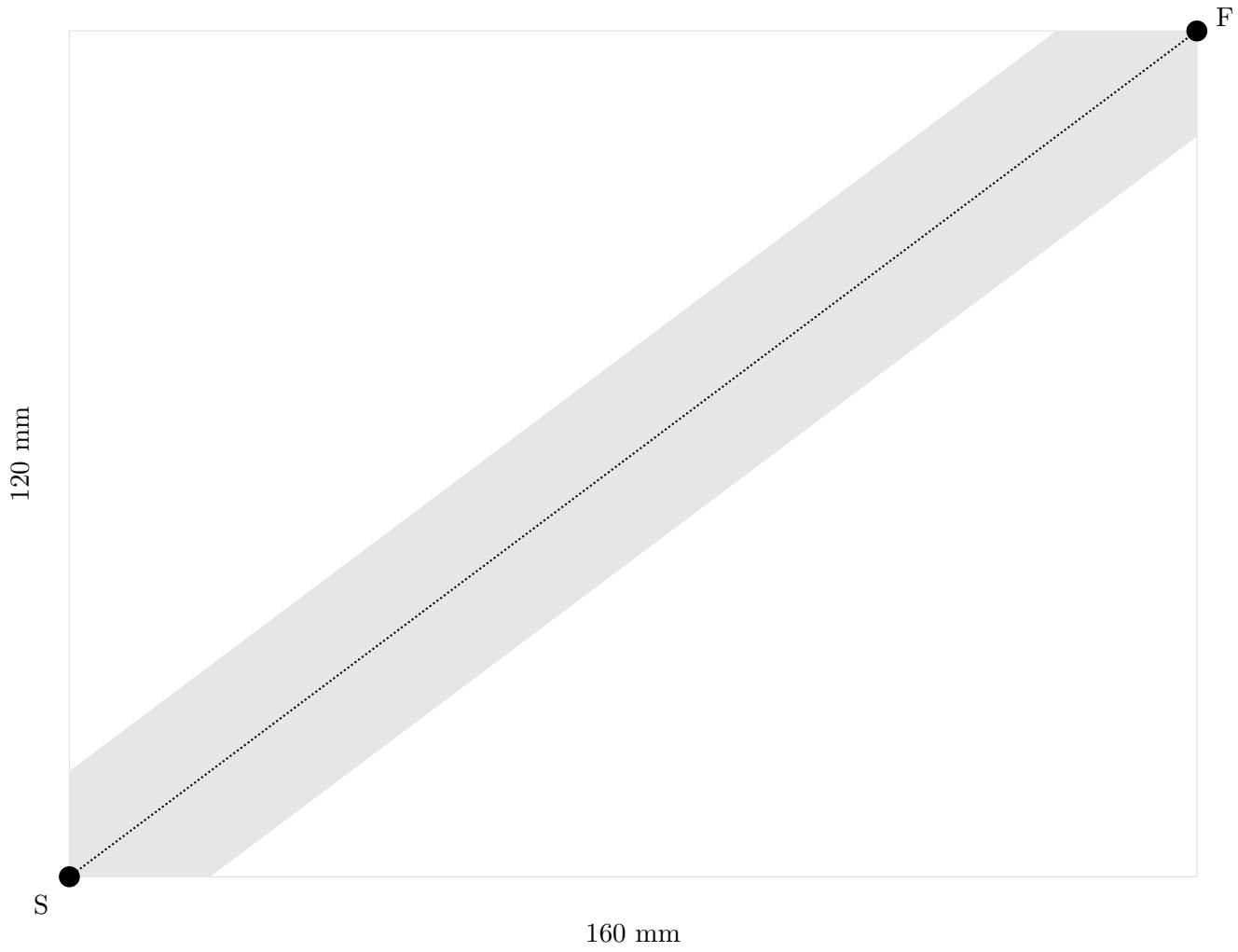
Problem 3: Make a staircase inside the gray strip, with a smaller greatest vertical gap from the diagonal than this path. Compare the two staircase lengths with the diagonal's length.



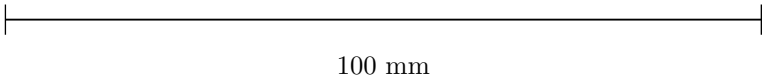
The straight diagonal is 200 mm long. The strip reaches 15 mm vertically above and below it.



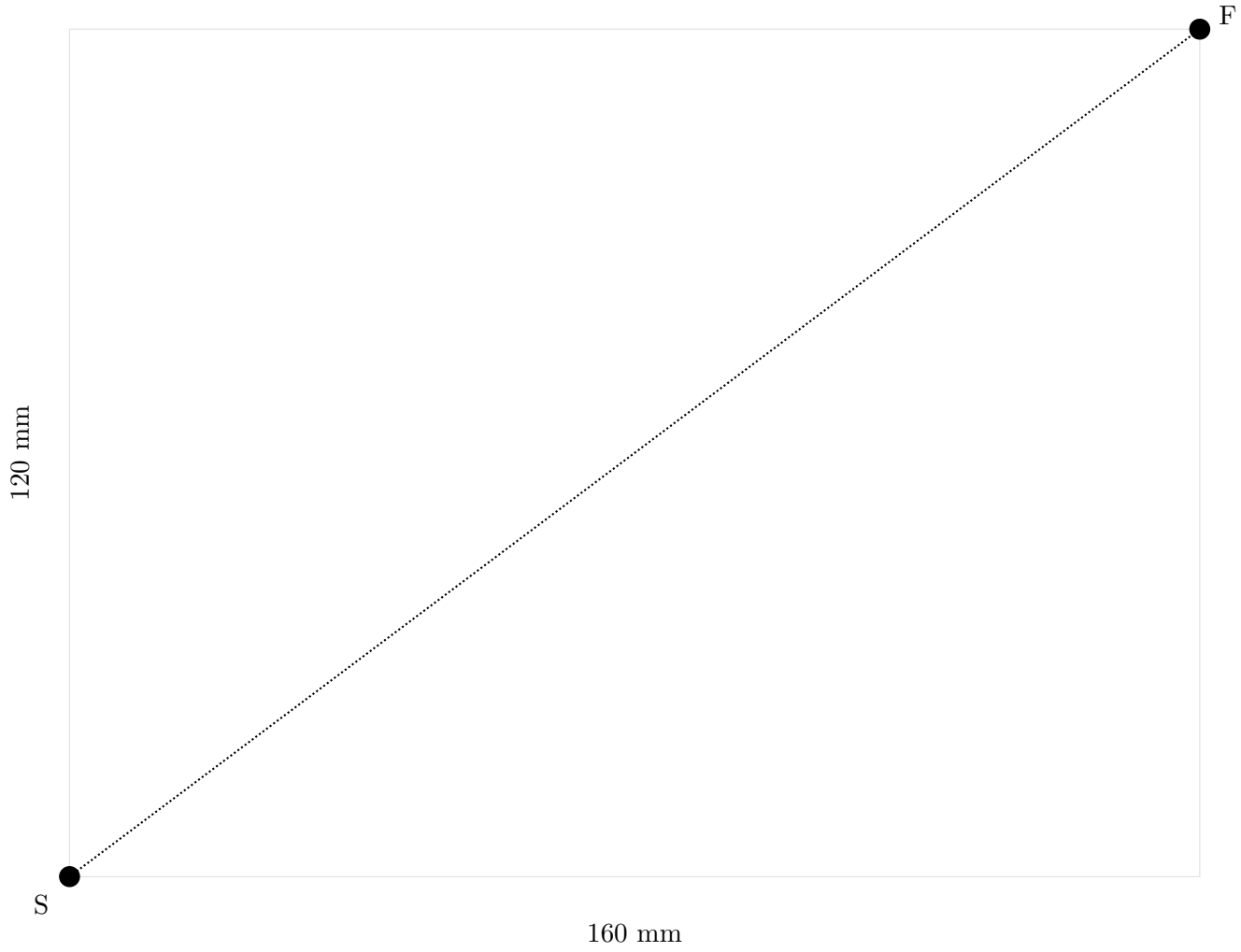
Problem 4: Find the fewest turns possible for an S-to-F staircase inside the gray strip. Its pieces may follow the strip's boundary. Explain why fewer turns cannot work.



The gray strip reaches 15 mm vertically above and below the dotted diagonal, clipped by the rectangle.



Problem 5: Allow leftward pieces as well as rightward and upward pieces. Make two very different S-to-F paths of length 360 mm. Could such a path be shorter than 280 mm? Explain your answer.



A straight diagonal is 200 mm long.

